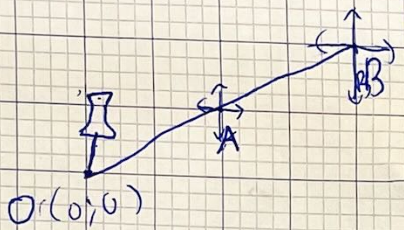
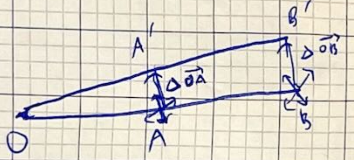
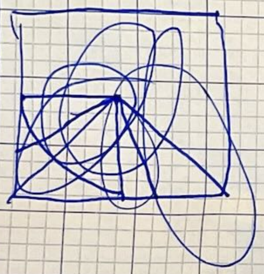
$$|\Delta \vec{OB}| = 2\,|\Delta \vec{OA}|$$

$$\frac{|\Delta \vec{OB}|}{|\Delta \vec{OA}|} = 2$$

$$\frac{|\Delta B|}{|\Delta A|} = 2$$







$$\vec{OB} = 2\vec{OA} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \vec{OA}$$

$$\Delta \vec{OB} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \Delta \vec{OA}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

$$\frac{\begin{pmatrix} 1 \\ 4 \end{pmatrix}}{\begin{pmatrix} 1 \\ 2 \end{pmatrix}} = ???$$

$$\begin{pmatrix} ax+by \\ cx+dy \end{pmatrix} = \cancel{\begin{pmatrix} f(x,y) \\ g(x,y) \end{pmatrix}}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

$$\begin{pmatrix} a+2b \\ c+2d \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

$a+2b=1$
$c+2d=4$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix} \forall x;y$$

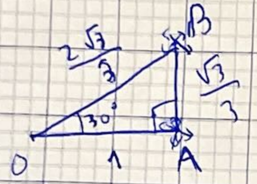
$$\begin{pmatrix} ax+by \\ cx+dy \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix} \forall x;y$$

$a = 2$
$b = 0$
$c = 0$
$d = 2$

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

futurebook

futurebook

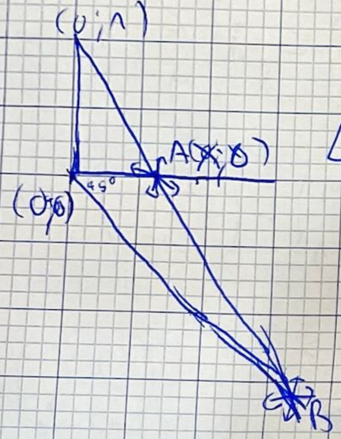


$$\Delta B = \begin{pmatrix} \frac{2\sqrt{3}}{3}\cos 30° & \frac{2\sqrt{3}}{3}\sin 30° \\ -\frac{2\sqrt{3}}{3}\sin 30° & \frac{2\sqrt{3}}{3}\cos 30° \end{pmatrix} \Delta A$$

$$|\Delta B| = \frac{2\sqrt{3}}{3} |\Delta A|$$

$$\frac{|\Delta B|}{|\Delta A|} = \frac{2\sqrt{3}}{3}$$

$\frac{|\Delta B|}{|\Delta A|} \neq C$



ΔA

ΔB

$A(x;0)$

$\exists B\left(\cancel{\frac{-1}{2}}\right)$

$\left(\frac{-1}{2}; \frac{-1-2x}{2x}\right)$

$(0;1)$

$(x;0)$

$(2x;-1)$

$(3x;-2)$

$(ax; -a-1)$

$ax = +a+1$

$\frac{x}{x-1}$

$$\frac{(x-1)\cdot 1 - x(1)}{(x-1)^2}$$

$$\frac{-1}{(x-1)^2}$$

$\frac{x}{x-1}$

$\frac{1}{x-1}$

$$\Delta B \neq \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Delta A$$

$A(x;0)$

$\exists B\left(\frac{x}{x-1}; \frac{-x}{x-1}\right)$

$a = \frac{1}{x-1}$

$a(x-1) = 1$

$|\Delta B|$

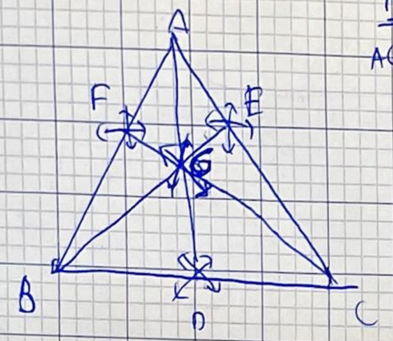
$|\Delta A| = |\Delta x|$

$$|\Delta B| = \sqrt{\left(\frac{x+\Delta x}{x+\Delta x-1} - \frac{x}{x-1}\right)^2}$$

$$|\Delta B| = \sqrt{2}\left(\frac{x+\Delta x}{x+\Delta x-1} - \frac{x}{x-1}\right)$$

$$\frac{|dB|}{|dA|} = \frac{-1}{(1_x-1)^2} = \frac{-1}{(|A|-1)^2}$$

$$\frac{dB}{dA} \overset{?}{=} \begin{pmatrix} \cos 45° & -\sin 45° \\ \sin 45° & \cos 45° \end{pmatrix} \frac{-1}{(|A|-1)^2}$$



$$\frac{AE}{AC-AE} = \frac{AF}{AB-AF}\,\frac{BD}{BC}$$

$$AE = \frac{AC\cdot AF\cdot BD}{(AB-AF)BC} - AE\left(\frac{AF\cdot BD}{(AB-AF)BC}\right)$$

$$\cancel{AE = AF\left(\frac{BD\,EC}{FB\cdot DC}\right)}$$

$$\cancel{|\Delta AE| = |\Delta AF|}$$

A ; B ; C ; D fixed

E ; F ; G

$$\frac{\Delta E}{\Delta F}$$

F

$|\Delta F|$ $\quad |\Delta E| = ?$

$$\frac{AE}{EC} = \frac{AF}{AB}\,\underbrace{\frac{BD}{DC}}_{\text{constant}}$$

$$\cancel{\frac{|\Delta E|}{|\Delta F|} = \frac{AC}{AB}}$$

$$\cancel{\frac{AE}{AC} = 1+\left(\frac{AF}{AB}-1\right)\frac{BD}{BC}}$$

$$\cancel{= 1+\frac{AF}{AB}\,\frac{BD}{BC}-\frac{BD}{DC}}$$

$$\frac{dD}{dE}\;\frac{dE}{dF}\;\frac{dF}{dD} = \begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}$$

(A ; B ; C) (F) (D) (E)

$$\boxed{\left(\frac{dD}{dE}_{(F)}\;\frac{dE}{dF}_{(D)}\;\frac{dF}{dD}_{(E)}\right)_{(A;B;C)} = \begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}}$$

$$\cancel{\frac{AE}{AC}} =$$

$AE = AF$

:)

$$AE = \frac{\frac{AC\cdot AF\cdot BD}{(AB-AF)DC}}{1+\frac{AF\cdot BD}{(AB-AF)DC}} \begin{matrix})\,M \\ )\,N\end{matrix}$$

$$\cancel{\frac{NM'-MN'}{N^2}}$$

$$\frac{NM'-MN'}{N^2}$$

$$\frac{|dE|}{|dF|}_{(A;B;C;D)} = \dots\dots$$

$$\cancel{M' = (-DC)(AC\cdot AF\cdot BD)}$$

$$1+\frac{AF\cdot BD}{FB\cdot DC}$$

$$\cancel{N' = (AB-AF}$$

$$\cancel{\left(1+\frac{AE}{EC}\right)}$$

$$N = \frac{AC}{EC}$$

$$M = \frac{AC\cdot AE}{EC}$$

$$\cancel{M' = BF\cdot DC\cdot AC\cdot BD + AC\cdot BD\cdot AF\cdot D}$$

$$M' = \frac{AC\cdot BD\cdot DC\cdot AB}{BF^2 DC^2}$$

$$\cancel{N' = BF\cdot BD\cdot DC + AF\cdot BD\cdot DC}$$

$$N' = \frac{AB\cdot BD\cdot DC}{BF^2 DC^2} = \frac{AB\cdot BD}{BF^2 DC}$$

$$M' = \frac{AB\cdot AC\cdot BD}{BF^2 DC}$$

$$\frac{dE}{dF}_{(A;B;C;D)}\;\frac{dF}{dD}_{(A;B;C;E)}\;\frac{dD}{dE}_{(A;B;C;F)} = \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}\cos 180^\circ & -\sin 180^\circ\\ \sin 180^\circ & \cos 180^\circ\end{pmatrix} = \begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}$$

$$\frac{NM'-MN'}{N^2}$$

$$\frac{\frac{AC}{EC}\,\frac{AB\cdot AC\cdot BD}{BF^2 DC} - \frac{AC\cdot AE}{EC}\,\frac{AB\cdot BD}{BF^2 DC}}{\frac{AC^2}{EC^2}}$$

$$\frac{\cancel{AB}\,AC\cdot BD\cdot EC}{BF^2\cdot DC\cdot \cancel{AC}} - \frac{AE\cdot AB\cdot BD\cdot EC}{BF^2 DC\cdot AC}$$

$$\frac{|dE|}{|dF|}_{(A;B;C;D)} = \frac{AB}{AC}\,\frac{BD}{DC}\left(\frac{EC}{BF}\right)^2$$

$$\frac{AB\cdot BD\cdot EC^2}{BF^2\cdot DC\cdot AC}$$

$$\Rightarrow \frac{|dE|}{|dF|}_{(A;B;C;D)}\;\frac{|dF|}{|dD|}_{(A;B;C;E)}\;\frac{|dD|}{|dE|}_{(A;B;C;F)} = 1$$